Hyperspectral Image Classification Based on Multibranch Adaptive Feature Fusion Network

Chen Li, Yi Wang, *Member, IEEE*, Zhice Fang, and Penglei Li

***Abstract*— Convolutional neural networks (CNNs) are widely used in hyperspectral image classification (HSIC) due to their exceptional performance. However, current methods for multiscale feature extraction typically rely on single-branch CNNs, potentially causing interference among features of varying scales. To mitigate this issue, we present a multibranch adaptive feature fusion network (MBAFFN) classification method. MBAFFN enhances feature uniqueness and improves the accuracy and reliability of classification results by extracting information at multiple scales through three parallel branches. Furthermore, to address the challenge of capturing global features within CNNs, we introduce a global detail attention (GDA) mechanism aimed at bolstering the network's capability to capture comprehensive information. In addition, we mitigate the issue of neglecting center-pixel importance in convolution operations through a distance suppression attention (DSA) design. To effectively integrate outcomes from multiple branches, we propose a pixel-based adaptive feature fusion strategy, thereby increasing the proportion of features conducive to improved classification results. Lastly, auxiliary loss functions are employed to train the multibranch network. Experimental results on four benchmark datasets demonstrate the superiority of our approach over several state-of-the-art methods, particularly in managing imbalanced small samples. Furthermore, ablation studies validate the effectiveness of the proposed modules.**

***Index Terms*— Convolutional neural networks (CNNs), feature fusion, hyperspectral image classification (HSIC), multibranch.**

I. Introduction

HYPERSPECTRAL images offer abundant spectral information, facilitating precise detection and differentiation of diverse land features. By integrating spectral and spatial information from remote sensing images, hyperspectral image classification (HSIC) techniques provide a comprehensive and accurate approach for land cover identification in various domains, including precision agriculture [1], urban development [2], and mineral exploration [3]. HSIC has long been a hot issue in remote sensing. Traditional machine learning methods [4], [5], [6] often emphasize feature extraction and classification algorithms but are subject to certain limitations. However, the high dimensionality, spectral confusion, and data sparsity of hyperspectral images may present challenges such as the curse of dimensionality, sample sparsity, and overfitting for traditional methods. Furthermore, these methods are less efficient in handling large-scale datasets and have limited capacity for addressing nonlinear classification problems.

In recent years, the advent of deep learning techniques [7] has led researchers to explore their application in HSIC, with the goal of improving classification accuracy. Deep learning-based methods not only allow for automatic feature learning from raw data, thereby overcoming the limitations of manual feature extraction but also exhibit stronger feature representation capabilities and model generalization abilities. As a result, deep learning-based approaches have attracted widespread attention in the field of HSIC. Chen et al. [8] proposed a spectral–spatial classification method based on deep belief networks, achieving improved classification accuracy compared to traditional methods. Mou et al. [9] treated each pixel of the hyperspectral image as an ordered sequence of data and employed recurrent neural networks to extract the internal sequential structural features of pixels, thereby improving classification accuracy. Hong et al. [10] developed a new model called SpectralFormer, a transformer-based backbone network, which effectively learns local spectral sequence information from adjacent spectral bands of hyperspectral remote sensing images and enhances the representation capability of spectral sequence information. Li et al. [11], [12] utilized features to construct a new remote sensing knowledge graph from scratch, supporting inference recognition of unseen remote sensing image scenes, addressing zero-shot and generalized zero-shot classification of remote sensing image scenes.

Hyperspectral remote sensing data is structured in a 3-D cube, and 3-D convolutional neural networks (CNNs) are commonly used in hyperspectral remote sensing image classification [13], [14], [15], [16] because of their ability to extract both spectral and spatial features concurrently. Zhong et al. [17] combined residual networks [18] with 3-D convolutional networks to create a deeper and more robust 3-D convolutional network. Wang et al. [19] enhanced this work by integrating dense connections with 3-D convolutional networks, allowing the network to learn correlations between features at different stages. However, relying solely on 3-D convolution for spectral–spatial feature extraction often results in over-smoothing and inadequate representation of spectral

Manuscript received 20 July 2024; accepted 19 August 2024. Date of publication 29 August 2024; date of current version 9 September 2024. This work was supported in part by the Joint Funds of the National Natural Science Foundation of China under Grant U21A2013, in part by the National Natural Science Foundation of China under Grant 61271408, and in part by the State Key Laboratory of Resources and Environmental Information System. *(Corresponding author: Yi Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3449878

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features. Roy et al. [20] suggested a hybrid convolutional network utilizing both 3-D and 2-D convolutions, adding 2-D convolutions after 3-D convolutions to improve spatial feature extraction. Li et al. [21] introduced a dual-branch network that combines 2-D and 3-D convolutions, incorporating an attention mechanism to enhance the model's performance. Also, various attention modules have been proposed, which is categorized into spatial domain attention [21], [22], [23], channel domain attention [24], [25], [26], [27], and mixed domain attention [28], [29], [30], [31], [32], [33]. Nikzad et al. [34] developed a lightweight and effective joint spectral–spatial attention mechanism to exploit the interplay between spectral bands and spatial features, successfully addressing the limitations of considering both spectral and spatial properties simultaneously in 1-D and 2-D CNNs. Kanthi et al. [35] introduced a novel 3-D attention network capable of capturing remote spatial dependencies in both horizontal and vertical directions of hyperspectral data, as well as saliency at different levels in various spectral bands.

Multiscale features play a crucial role in distinguishing complex land cover distributions in real-world settings. Xu et al. [36] integrated convolutional features from different stages to extract spatial features at multiple scales. Cao and Guo [37] introduced a cascaded dual-scale cross network that automatically captures spectral–spatial features of hyperspectral images across different scales by employing consecutive dual-scale cross blocks with varied convolution kernel sizes. Yang et al. [38] implemented a spectral cascaded expansion network for extracting spectral features, utilizing parallel multipath networks to capture features at different scales and achieving multipath feature fusion through fusion layers. To holistically address spectral and spatial characteristics, Yang et al. [39] further proposed integrating spectral information online when extracting multiscale spatial features. Cai et al. [40] utilized three branches with different-sized convolution kernels to extract features at various scales, merging these features via a Bole convolution layer to achieve feature fusion. Meanwhile, Zhong et al. [41] proposed a parallel multi-input mechanism convolutional network to classify features across different scales by extracting them through four parallel networks and conducting feature fusion through multiple fully connected layers after multiscale feature extraction.

Although the studies described above have achieved promising performance, there are still three significant issues that remain to be solved. First, the current multiscale classification frameworks often neglect interference among different scales during feature extraction, leading to significant discrepancies in how instances from the same class are represented in feature space. Consequently, this affects the decision boundary and the stability of classifiers. Second, CNN-based classification models insufficiently consider the importance of central pixels, which could lead to a decrease in the model's performance with regard to spatial invariance. In addition, CNNs struggle to capture global information effectively, leading to limited contextual understanding when dealing with complex scenes or large-scale objects. Third, the simultaneous integration of multiple branch results remains a persistent challenge in HSIC. Previous approaches primarily relied on fully connected or convolutional layers for fusion, compromising both model flexibility and interpretability.

To overcome the difficulties described above, we propose a multibranch adaptive feature fusion network (MBAFFN) method for HSIC. As shown in Fig. 1, we employ three parallel branch networks to extract information at different scales, effectively mitigating feature interference issues. Each branch combines 2-D and 3-D convolutions to concurrently extract spectral and spatial features, fully leveraging the 3-D nature of hyperspectral data and avoiding the process of information separation and reintegration. In addition, we design global detail attention (GDA) and distance suppression attention (DSA) modules for each branch to address the deficiencies of CNNs in extracting global features and to enhance the focus of convolutional kernels on central pixels. Finally, we propose an adaptive feature fusion strategy to integrate the results of multiple branches more effectively. To better train the multibranch network, we employ auxiliary losses (ALs).

The main contributions of this article are summarized as follows.

1) To mitigate feature interference when capturing multiscale features using CNNs, we introduce a novel MBAFFN for HSIC. This strategy effectively ensures the uniqueness and representativeness of features at different scales.
2) To enhance the CNNs capability to capture global information and its focus on central pixels, we propose the GDA and DSA modules to enhance the convolutional network's extraction of global features and focus on central pixels.
3) To make full use of the multiscale information extracted from each branch, we propose an efficient adaptive feature fusion strategy to integrate the features from multiple branches and suppress the interference features.

The rest of this article is organized as follows. Section II provides a detailed description of the proposed framework. Section III presents extensive experiments and offers a detailed analysis of the classification results. Section IV concludes the article.

II. Methodology

The flowchart of this study is shown in Fig. 1. This framework consists of four primary components: data grouping, extraction of multiscale spectral features, extraction of multiscale spatial features, and adaptive feature fusion. To improve the training of subnetworks, ALs are incorporated for each branch, and a weighted summation is used to calculate the overall loss. First, we divide the hyperspectral image data into three groups along the spectral dimension and process the grouped data through feature extraction networks operating at three different scales. Next, the framework enhances the feature extraction process by integrating global information via the GDA module and emphasizing the significance of central pixels through the DSA module. Subsequently, the multiscale features extracted by the network are input into the adaptive feature fusion module to derive classification features. Finally, the integrated multiscale features are input into the softmax classifier for classification.

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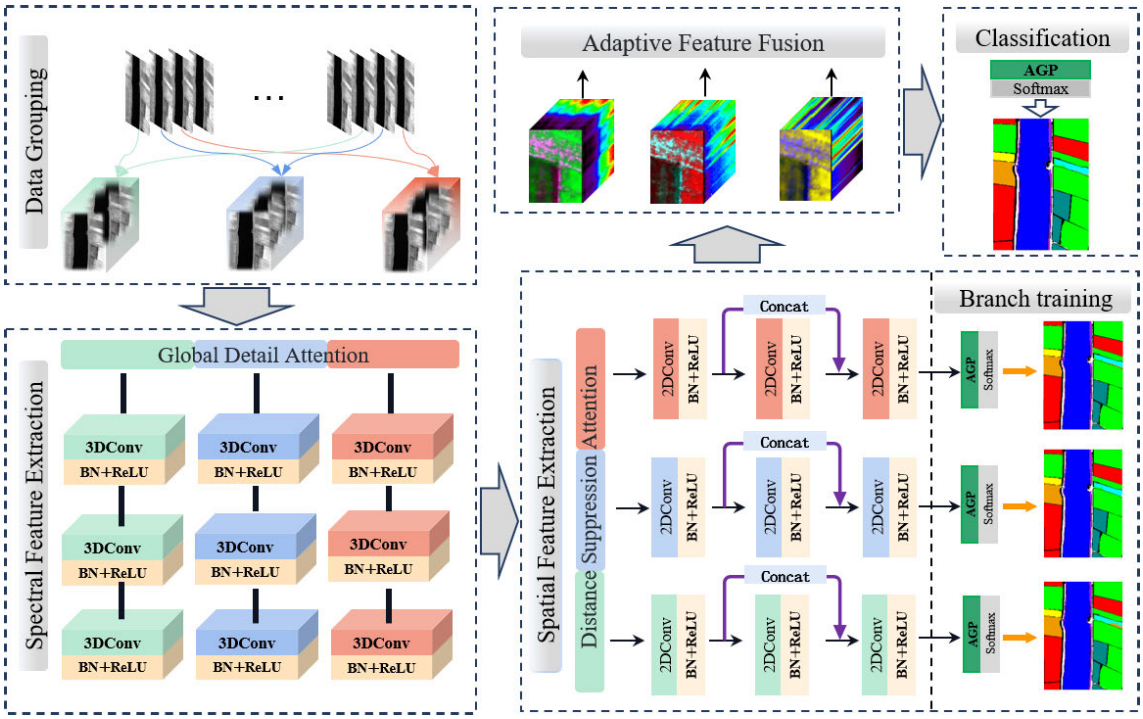


Fig. 1. Proposed MBAFFN. Layers with the same colors and shapes represent corresponding operations.

A. Grouping Strategy

Hyperspectral data exhibits a large number of features. As the dimensionality increases, the data becomes sparser in the feature space, resulting in the distances between data points to increase, which brings challenges for distinguishing samples from different classes. Some existing methods [43] prioritize spectral dimension grouping strategies but overlook the coherent integration of spatial and spectral information. To address this, we propose a new grouping strategy tailored for multibranch networks. This method reduces redundant inputs, thereby lowering computational complexity while ensuring that each branch can extract both spectral and spatial features simultaneously, as shown in Fig. 2. In this study, we apply different grouping strategies to publicly available hyperspectral datasets. Specifically, the Indian Pines (IP) dataset and the Pavia University (PU) dataset undergo a single round of grouping, resulting in the following input representations for the branches: $X^1_{\text{one}}, X^1_{\text{two}}, X^1_{\text{three}} \in R^{W\times H\times D/3}$, where $X \in R^{W\times H\times D}$ represent the original data of a given hyperspectral image. Two rounds of data grouping are performed on the Longkou dataset and Hanchuan dataset due to their high spectral dimensions and spatial resolution, and redundant information exists in adjacent bands. Thus, the average of three adjacent bands $X_{\text{avg}} \in R^{W\times H\times D/3}$ is computed. Following two rounds of grouping, the final input for each branch becomes: $X^2_{\text{one}}, X^2_{\text{two}}, X^2_{\text{three}} \in R^{W\times H\times D/6}$. The grouping and fusion process can be expressed as follows:

$$X^1_{\text{one}}, X^1_{\text{two}}, X^1_{\text{three}} = G(X) \tag{1}$$

$$X_{\text{avg}} = F\left(X^1_{\text{one}}, X^1_{\text{two}}, X^1_{\text{three}}\right) \tag{2}$$

$$X^2_{\text{one}}, X^2_{\text{two}}, X^2_{\text{three}} = G\left(X_{\text{avg}}\right). \tag{3}$$

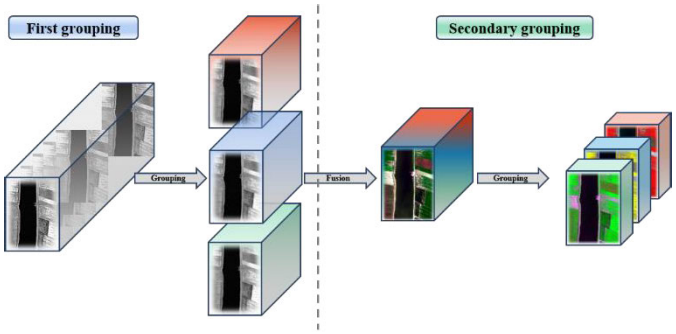


Fig. 2. Grouping process.

B. Multiscale Feature Extraction

Multiscale features play a vital role in the classification of hyperspectral images. Hyperspectral images contain various spatial information, and incorporating multiscale features enables the capture of spatial information at different scales. Moreover, the extraction of multiscale features also improves the robustness of classification methods against noise interference. To address feature interference in multiscale feature extraction using traditional CNNs [36], we employ a multibranch parallel network and utilize convolution kernels of various sizes for effective feature extraction. Specifically, the network comprises three branches, and the multiscale feature extraction of each branch consists of two stages.

1) Spectral Feature Extraction: The spectral feature extraction module consists of three layers of 3-D convolution, as listed in Table I. From the table, the input and output of the first and last layers are fixed to 1, while the number of convolution kernels in the middle layer is equivalent to the size of the kernel, ensuring the diversity of feature extraction.

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The overall network structure is diamond-shaped, which fully exploits the advantages of hybrid convolution.

2) Spatial Feature Extraction: Similarly, it utilizes a three-layer network convolution, with a slightly different number of convolution layers for different experimental sets. The specific details of the multiscale feature network are listed in Table I. From the table, the spatial size remains fixed at 11×11. D^{dataset} represents the spectral dimension of the input data after grouping, and $D^{\text{dataset}} \in \{D^{\text{IP}} = 66, D^{\text{PU}} = 34, D^{\text{LK}} = D^{\text{HC}} = 30\}$, K_{scale} represents the kernel size, with three different scales of 7, 5, and 3, $N_{\text{spa}}^{\text{dataset}}$ represents the number of convolution kernels of the spatial feature extraction stage, and $N_{\text{spa}}^{\text{dataset}} \in \{N_{\text{spa}}^{\text{IP}} = 56, N_{\text{spa}}^{\text{LK}} = 40, N_{\text{spa}}^{\text{PU}} = N_{\text{spa}}^{\text{HC}} = 48\}$.

C. Global Detail Attention

The spectral information in each band of the image provides insight into the unique characteristics of the object. By incorporating spectral attention mechanisms, the model can autonomously learn and focus attention on crucial spectral bands while disregarding those less relevant to the classification task. Considering the limitations of capturing long-range dependencies and global information through convolution operations, we propose a GDA, inspired by the concept of the compressed attention mechanism [44], as shown in Fig. 3. This module consists of three branches: the middle branch dedicated to learning detailed information for the central pixel; the bottom branch for compressing global spatial information related to spectral attention; and the top branch for compressing global spectral information to facilitate spectral attention. In the subsequent sections, each of these branches will be thoroughly discussed.

1) The middle branch is used to extract detailed information from the central pixel. Consider an input patch size $x \in R^{w \times h \times d}$, where w and h represent the spatial dimensions in the horizontal and vertical directions of the patch, and d represents the spectral dimension. Initially, the central pixel of the input data denoted as $x_{\text{centre}} \in R^{1 \times 1 \times d}$ is extracted. Then, the pixel undergoes three layers of 3-D convolution with batch normalization applied after each convolution layer, where the convolution kernel size is set to (1, 1, 7). To enhance the network's capacity for learning more comprehensive feature representations, the middle layer is assigned a size of 7, while the last layer aggregates the previously learned features into a size of 1. With this network structure, the risk of overfitting is effectively reduced, rendering the network more adaptable and efficient. Finally, the spectral feature vector X_{centre} representing localized details is obtained. The mathematical expression for this specific process is given as follows:

$$X_{\text{centre}} = 3\text{DConv}(3\text{DConv}(3\text{DConve}(x_{\text{centre}}, \omega, \sigma))) \tag{4}$$

where ω and σ represent the coefficient matrix and the bias, respectively, during the convolution process.

2) The bottom branch compresses global spatial information by performing global adaptive pooling on the input data x. This operation reduces the spatial dimension, yielding a 1-D spectral vector $x_{\text{gap}} \in R^{1 \times 1 \times d}$ that contains the global spatial information

$$x_{\text{gap}} = \frac{1}{w \times h \times d} \sum_{i=1}^{w} \sum_{j=1}^{h} \sum_{k=1}^{d} x_{i,j,k} \tag{5}$$

where i, j represent the spatial position of the pixel, and k represents the channel dimension of the pixel. Then, a convolution operation similar to the one used for extracting central spectral features is performed on x_{gap} to extract a feature vector X_{Gspa} containing global spatial information. The specific process can be expressed as follows:

$$X_{\text{Gspa}} = 3\text{DConv}\big(3\text{DConv}\big(3\text{DConve}\big(x_{\text{gap}}, \omega, \sigma\big)\big)\big) \tag{6}$$

where ω and σ represent the coefficient matrix and the bias, respectively, during the convolution process.

The next step involves multiplying X_{centre} by X_{Gspa}, which contains the compressed global spatial information. This multiplication is followed by applying a softmax function to derive the fused feature vector X_{CGspe}. By combining the central detailed information and global spatial information, X_{CGspe} can be more effectively fused with the global spectral feature vector X_{Gspe} extracted from the top branch. The specific process can be expressed as follows:

$$X_{\text{CGspa}} = \text{Softmax}\big(X_{\text{centre}} * X_{\text{Gspa}}\big). \tag{7}$$

3) The top branch aims to extract global spectral information. First, the input data x undergoes a 3-D convolution operation with a kernel size of (1, 1, d) to compress each pixel along the spectral dimension, resulting in d feature maps denoted as $f_i, i \in 1, 2, \ldots, d$. Then, these feature maps are reshaped to match the dimensions of the input data, $F \in R^{w \times h \times d}$. They are further utilized to extract global spectral features using 3-D convolution with a kernel size of (1, 1, 7), which yields a spectral feature vector containing comprehensive global spectral information. The entire process is mathematically represented as follows:

$$f_i = \text{Bn}(3\text{DConv}(x, \omega, \sigma)) \tag{8}$$

$$F = \text{Reshape}(f_1, f_2, \ldots, f_d) \tag{9}$$

$$X_{\text{Gspe}} = \text{Bn}(3\text{DConv}(F, \omega, \sigma)) \tag{10}$$

where ω and σ represent the coefficient matrix and the bias, respectively, employed in the convolution process.

After that, the resulting global spectral feature X_{Gspe} is combined with X_{CGspa} containing central pixel details information and global spatial feature to yield X_{merge}. Finally, X_{merge} is subjected to a Sigmoid function, producing the final attention weight coefficient matrix of att. The complete procedure is summarized as follows:

$$X_{\text{merge}} = X_{\text{CGspa}} * X_{\text{Gspe}} + X_{\text{Gspe}} \tag{11}$$

$$\text{att} = \text{Sigmoid}\big(X_{\text{merge}}\big). \tag{12}$$

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TABLE I
DETAILED DATA-FLOW OF MULTISCALE FEATURE EXTRACTION NETWORK

	Layer	Input size	Output size	Kernel size	Kernel Number
Spectral CNN	3DConve	$(11,11,D^{dataset})$	$(11,11,D^{dataset})$	$(1,1,K_{scale})$	K_{scale}
	3DConve	$(11,11,D^{dataset})$	$(11,11,D^{dataset})$	$(1,1,K_{scale})$	K_{scale}
	3DConve	$(11,11,D^{dataset})$	$(11,11,D^{dataset})$	$(1,1,K_{scale})$	1
Spatial CNN	2Dconve	$(11,11,D^{dataset})$	$(11,11,N_{spa}^{dataset})$	(K_{scale},K_{scale})	$N_{spa}^{dataset}$
	2Dconve	$(11,11,N_{spa}^{dataset})$	$(11,11,N_{spa}^{dataset})$	(K_{scale},K_{scale})	$N_{spa}^{dataset}$
	2Dconve	$(11,11,2N_{spa}^{dataset})$	$(11,11,N_{spa}^{dataset})$	(K_{scale},K_{scale})	$N_{spa}^{dataset}$

ANNOTATION: $D^{dataset}$ represents the spectral dimension of the input data for different datasets. K_{scale} represents the size of the convolutional kernel. $N_{spa}^{dataset}$ represents the number of kernels in the spatial feature extraction stage.

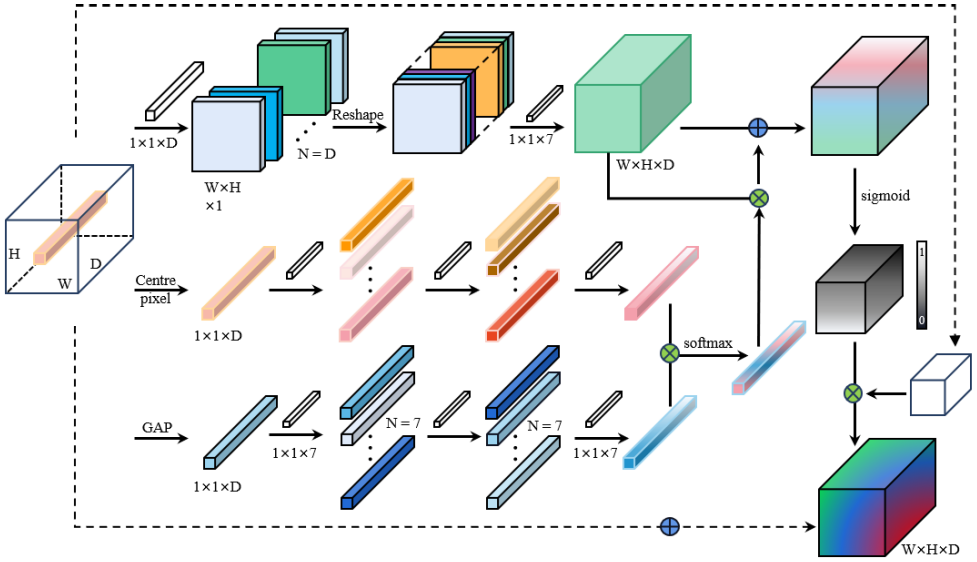


Fig. 3. GDA. Three branches are used to learn the attention of the central pixel, compress the global spatial information of the spectral attention, and compress the global spectral information of the spectral attention, respectively.

Finally, we integrate the attention mechanism with the input data x. To avoid the potential loss of crucial feature information that may occur when directly multiplying the attention with the input data, we employ a combination of multiplication and addition to merge the attention into the original data. This process produces the attention-enhanced original data of x_{att}. The entire process is represented as follows:

$$x_{\text{att}} = x * \text{att} + x. \tag{13}$$

D. Distance Suppression Attention

In HSIC, pixel categories are influenced by both their spectral features and spatial relationships with neighboring pixels. However, CNNs often extract features from neighboring pixels during classification without considering the importance of the central pixel. According to the first law of geography [42], the distance between geographic features is the most direct indicator of their similarity. To enhance the influence of the central pixel on classification, we calculate the similarity between the neighborhood pixels $x_n \in R^{1\times\sim1\times\sim d}$, $n \in \{1, 2, \ldots, w*h-1\}$, and the central pixel x_{centre} based on their distances. Furthermore, we propose a DSA module that computes a distance mapping $D(x_{\text{centre}}, x_n)$ by evaluating the Euclidean distance

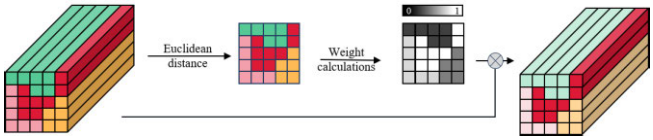


Fig. 4. Distance suppression attention.

from each surrounding pixel to the central pixel, as shown in Fig. 4. The distance mapping can be computed as follows:

$$D(x_{\text{centre}}, x_n) = \sqrt{\sum_{k=1}^{d} \left(x_{\text{centre}}^k - x_n^k\right)^2} \tag{14}$$

where x_{centre}^k and x_n^k represent the kth dimension of the central pixel and the neighboring pixel, respectively.

Utilizing distance-based similarity alone is inadequate for creating a direct weight matrix for attention, as closer geographic features exhibit greater spatial similarity. For example, the distance between a pixel and itself is 0. In this study, we develop a transformation function that study, maps distance values to a numerical range from 0 to 1. As the mapped numerical value decreases, the weight tends to approach 1, as shown in Fig. 5. In the coordinate system, the maximum value of 1 is obtained when $x = 0$, and the decreasing y-values toward the edges indicate a decreasing similarity. In addition,

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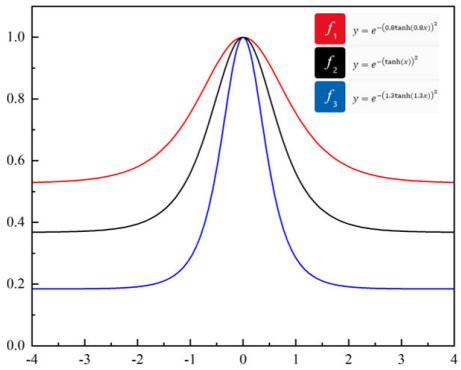


Fig. 5. Custom activation function.

the function prevents weights from reaching 0. This avoids information loss caused by long distances. We define the transformation function as follows:

$$f(x) = e^{-(b(\tanh(ax)))^2} \tag{15}$$

where a and b control the slope and magnitude of the function, respectively, and their values are obtained during training.

During network training, appropriate values for a and b are determined to ensure the generation of suitable spatial attention weight coefficients. In contrast to traditional spatial attention mechanisms that amplify signals, we prioritize the significance of central pixels by suppressing nonhomogeneous pixel values in the neighborhood, thereby enhancing the robustness of the network.

E. Adaptive Fusion Strategy and Loss Design

A reasonable feature fusion strategy can effectively fuse features from different scales and improve classification accuracy. Commonly used methods for multiscale feature fusion, such as element-wise addition or concatenation, assume equal importance for features from each scale. However, this assumption may overlook the complex characteristics of multiscale features in hyperspectral images. Some methods involve flattening features from different scales, concatenating them, and then using the sigmoid function to determine the weights. The sigmoid function may be saturated when the dimensions of multiscale features are too large or the input values are extremely large or extremely small. This study introduces a pixel-based adaptive fusion method, as depicted in Fig. 6. Based on the importance of corresponding feature points, we calculate the respective weights of W_{large}, W_{middle}, and W_{small} based on the significance of corresponding feature points. We multiply the weight coefficients by the corresponding features to obtain the fused feature of f_{merge}. The specific process can be represented as follows:

$$W_{\text{large}} = \frac{e^{f_{\text{large}}}}{e^{f_{\text{large}}} + e^{f_{\text{middle}}} + e^{f_{\text{small}}}} \tag{16}$$

$$W_{\text{middle}} = \frac{e^{f_{\text{middle}}}}{e^{f_{\text{large}}} + e^{f_{\text{middle}}} + e^{f_{\text{small}}}} \tag{17}$$

$$W_{\text{small}} = \frac{e^{f_{\text{small}}}}{e^{f_{\text{large}}} + e^{f_{\text{middle}}} + e^{f_{\text{small}}}} \tag{18}$$

$$f_{\text{merge}} = W_{\text{large}} * f_{\text{large}} + W_{\text{middle}} * f_{\text{middle}} + W_{\text{small}} * f_{\text{small}} \tag{19}$$

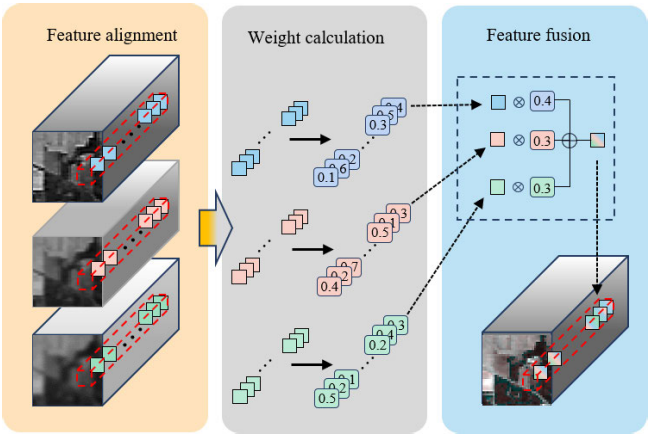


Fig. 6. Adaptive feature fusion strategy.

where f_{large}, f_{middle}, and f_{small} represent the extracted large-scale features, medium-scale features, and small-scale features obtained from three branches of the network, respectively. The proposed pixel-based multiscale adaptive feature fusion method not only integrates information from different scales in a rational and meticulous manner but also addresses the issue of feature loss caused by scale differences, thereby significantly improving classification accuracy.

Inspired by [31], we enhance each branch network by incorporating auxiliary loss functions to improve the training process and prevent bias toward a specific branch during backpropagation. Unlike the work in [31], our method assigns varying weights to the three auxiliary loss functions of L^{one}, L^{two}, and L^{three}, as well as the primary joint loss function of L^{joint}, to emphasize the importance of the joint primary loss. Specifically, we set the coefficient for L^{joint} as $\lambda^0 = 0.25$, while the coefficients for the other three branches are $\lambda^1 = 0.25$, $\lambda^2 = 0.25$, and $\lambda^3 = 0.25$. Details on coefficient determination are provided in Section III-F. The complete loss function of L is defined as follows:

$$\begin{aligned} L &= \lambda^0 L^{\text{joint}} + \lambda^1 L^{\text{one}} + \lambda^2 L^{\text{two}} + \lambda^3 L^{\text{three}} \\ &= -\frac{\lambda^0}{m}\sum_{i=1}^{m}\left[y_i \log\left(\hat{y}_i^{\text{joint}}\right) + (1-y_i)\log\left(1-\hat{y}_i^{\text{joint}}\right)\right] \\ &\quad -\frac{\lambda^1}{m}\sum_{i=1}^{m}\left[y_i \log\left(\hat{y}_i^{\text{one}}\right) + (1-y_i)\log\left(1-\hat{y}_i^{\text{one}}\right)\right] \\ &\quad -\frac{\lambda^2}{m}\sum_{i=1}^{m}\left[y_i \log\left(\hat{y}_i^{\text{two}}\right) + (1-y_i)\log\left(1-\hat{y}_i^{\text{two}}\right)\right] \\ &\quad -\frac{\lambda^3}{m}\sum_{i=1}^{m}\left[y_i \log\left(\hat{y}_i^{\text{three}}\right) + (1-y_i)\log\left(1-\hat{y}_i^{\text{three}}\right)\right] \end{aligned} \tag{20}$$

where $\hat{y}_i^{\text{joint}}$, $\hat{y}_i^{\text{one}}$, $\hat{y}_i^{\text{two}}$, and $\hat{y}_i^{\text{three}}$ are the corresponding predicted labels for the ith training sample, y_i is the true label, and m is the size of the training set.

III. Experiments and Analysis

This section first presents the datasets and model parameter settings used in the experiments. Then, it demonstrates the effectiveness of the proposed module through ablation experiments and compares it with other state-of-the-art classification

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methods. After that, it analyzes the impact of the parameters of the proposed method on the classification results, followed by an evaluation of the computational efficiency of all classification methods.

A. Experimental Settings

In this section, we select several typical CNN methods, including spectral–spatial residual network (SSRN) [17], fast dense spectral–spatial convolution network (FDSSC) [19], double-branch dual-attention mechanism network (DBDAMN) [27], multiattention fusion network (MAFN) [45], spectral–spatial unified network (SSUN) [36], and enhanced multiscale feature fusion network (EMFFN) [39], which include three of the latest methods: Bole Convolution with Three-Direction Attention Mechanism Network (BTA-Net) [40], a Three Branch Model (Tri-CNN) [46], and the center-to-surrounding interactive learning framework (CSIL) [47].

1) SSRN employs a multilayer 3-D convolutional residual structure for HSIC. The input size is $7 \times 7 \times d$, with a batch size of 16. The RMSProp algorithm was used for network training with a learning rate of 0.0003 for a total of 200 epochs.
2) FDSSC employs dense connections to fuse and extract features of different scales at each stage of 3-D convolution, improving the network's fitting speed. The input size is $9 \times 9 \times d$, with a batch size of 32. The RMSProp algorithm was used for network training with a learning rate of 0.0003 and 80 epochs.
3) DBDAMN extracts spectral and spatial features through different branches and refines and optimizes these features using various attention mechanisms. The input size is $9 \times 9 \times d$, with a batch size of 16. The Adam algorithm was used for network training, with a learning rate of 0.0003 and a total of 200 epochs.
4) MAFN acquires complementary information from different levels and scales, achieving feature reuse through multiple attention mechanisms and multilevel fusion. The input size is $7 \times 7 \times d$ with a batch size of 128 for the spectral branch of the network, and $27 \times 27 \times 30$ for the spatial branch. The Adam algorithm was used for network training with a learning rate of 0.001 and a total of 200 epochs.
5) SSUN is a dual-branch network designed for spectral and spatial domains. The spatial branch incorporates multiscale information. The input patch size for the spatial branch is set to 28×28, with PCA assigned a value of 4. The batch size is 64, and the network was trained using the Adam algorithm with a learning rate of 0.001 for a total of 500 epochs.
6) EMFFN employs dilated convolutions of various scales to extract spectral features in the spectral branch and different sizes of convolutional kernels to extract spatial features in the spatial branch. The input patch size for the spatial branch is set to 25×25, and PCA is assigned a value of 4. A batch size of 32 was used, and the network was trained with the RMSProp algorithm at a learning rate of 0.001 for 200 epochs.
7) BTA-Net enhances traditional convolutions with Bole convolutions and introduces a three-direction attention mechanism to mitigate small sample and sample imbalance issues. The spatial size is 16×16, and PCA is assigned a value of 30. The batch size of 256 was chosen and the network is trained using the Adam algorithm with a learning rate of 0.001 for 500 epochs.
8) Tri-CNN extracts spectral, spatial, and spectral–spatial features via three parallel branch networks. The input size is $11 \times 11 \times d$ with a batch size of 16. The Adam algorithm was used for network training with a learning rate of 0.001, and 100 epochs.
9) CSIL utilizes transformer with a hierarchical region sampling strategy to extract features of different scales. The input size is $27 \times 27 \times d$ and PCA is set to 5. The Adam algorithm was used for network training with a learning rate of 0.001 and 45 epochs.
10) MBAFFN has an input size of $11 \times 11 \times d$ for each branch of the network. The network is trained with a batch size of 256 and the Adam algorithm using a learning rate of 0.001 for 200 epochs, with early stopping applied at 40.

All the mentioned HSIC methods were implemented in PyTorch and the experiments were performed on a computer equipped with an Intel[1] Xeon[1] Gold 6226R processor and NVIDIA GeForce RTX 3090 GPU. Validation was carried out on the four datasets, with each experiment using both random and nonuniform sampling, and the average value was calculated over ten trials. To quantitatively evaluate the classification methods, three evaluation metrics, namely overall accuracy (OA), average accuracy (AA), and Kappa coefficient, were adopted in this study.

B. Dataset Description

In this study, as shown in Fig. 7 four publicly available datasets (IP, PU, Longkou, and Hanchuan) were chosen for experimentation. To ensure the proper functioning of all classification methods, the last two datasets with larger data volumes were spatially subsampled using a grid-based down-sampling strategy to reduce the spatial resolution by a quarter, while keeping the spectral dimension unchanged.

The IP dataset was acquired using the Airborne Visible/Infrared Imaging Spectrometer sensor in northwest Indiana. The image size is 145×145 with a spatial resolution of 20 m. The spectral range spans from 400 to 2500 nm. After removing water absorption and reducing noise, 200 spectral bands were preserved. The study area encompasses 16 land cover types, including corn, oats, wheat, woods, and others. We randomly selected 5%, 5%, and 90% samples for each class for training, validation, and testing, as listed in Table II.

The PU dataset was obtained by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over PU in northern Italy. The image size is 610×340 pixels, with a spatial resolution of 1.3 m, and comprises 103 spectral bands ranging from 430 to 860 nm. The study area encompasses a total of nine land cover types, including Asphalt, Meadows,

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TABLE II
LAND COVER TYPES AND NUMBER OF TRAIN/VAL/TEST/ SAMPLES IN THE FOUR DATASETS

IP				PU				LK				HC			
No.	Class	T/V	Test	Class	T/V	Test		Class	T/V	Test		Class	T/V	Test	
1	Alfalfa	2	42	Asphalt	66	6499		Corn	81	8436		Strawberry	112	10984	
2	Corn-n	71	1286	Meadows	186	18277		Cotton	21	2065		Cowpea	57	5550	
3	Corn-m	41	748	Gravel	21	2057		Sesame	8	748		Soybean	25	2520	
4	Corn	12	213	Trees	31	3002		Broad-l s	157	15419		Sorghum	13	1317	
5	Grass-p	24	435	Metal s	13	1319		Narrow-l s	10	1010		Water spinach	3	295	
6	Grass-t	37	656	Bare soil	50	4929		Rice	29	2899		Watermelon	11	1092	
7	Grass-p-m	1	26	Bitumen	13	1304		Water	168	16423		Greens	15	1453	
8	Hay-w	24	430	Bricks	37	3608		Roads and h	18	1730		Trees	45	4399	
9	Oats	1	18	Shadows	10	927		Mixed weed	13	1266		Grass	23	2313	
10	Soybean-n	49	874	Total	427	41922		Total	505	49996		Red roof	27	2605	
11	Soybean-m	123	2209									Gray roof	42	4121	
12	Soybean-c	30	533									Plastic	9	915	
13	Wheat	10	185									Bare soil	23	2233	
14	Woods	63	1139									Road	46	4522	
15	Buildings-G	19	348									Bright object	3	278	
16	Stone-S-T	5	83									Water	189	18401	
	Total	512	9225									Total	643	62998	

Gravel, and others. We randomly selected 1%, 1%, and 98% samples for each class for training, validation, and testing, as listed in Table II.

The Longkou (LK) dataset was obtained in Longkou, Hubei province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. The image size is 550 × 400 pixels (resampled to 275 × 200 pixels after spatial sampling), with a spatial resolution of approximately 0.463 m, comprising 270 bands covering a spectral range from 400 to 1000 nm. The study area represents a basic agricultural scene and encompasses six crops: Corn, Cotton, Sesame, Broadleaf Soybean, Narrowleaf Soybean, and Rice. We randomly selected 0.5%, 0.5%, and 99% samples for each class for training, validation, and testing, as listed in Table II.

The Hanchuan (HC) dataset was acquired in Hanchuan, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform. The image size is 1217 × 303 pixels (resampled to 608 × 151 pixels after spatial sampling) and has a spatial resolution of approximately 0.109 m, encompassing 274 bands covering a spectral range from 400 to 1000 nm. The study area represents a peri-urban region featuring a mix of buildings, water bodies, and agricultural land, including seven crops: Strawberries, Cowpeas, Soybeans, Sorghum, Water Spinach, Watermelon, and Chinese Cabbage, totaling 16 land cover types. We randomly selected 1%, 1%, and 98% samples for each class for training, validation, and testing, as listed in Table II.

C. Effects of Multiple Branches

For a better representation of the distribution of different classes in the feature space, Fig. 8 presents visualized results of classification accuracy using different strategies on four datasets. In Fig. 8(a)–(d), all the single-scale classification results of the IP dataset exhibit misclassification phenomena at various scales for Soybean-n, Soybean-m, and Corn-n. However, Fig. 8(d) mitigates the misclassification of neighboring land cover types by incorporating features from different scales. Fig. 8(e)–(h) shows the classification results of each branch on the PU dataset, where Trees, Bricks, and Shadows display more instances of misclassification due to their complex shapes and scattered distribution. Fig. 8(h) demonstrates improved classification accuracy for complex land cover types after incorporating multiscale features. In Fig. 8(i)–(l), it is evident that single-scale features are unable to distinguish Roads and Houses in the LK dataset, as well as elongated objects like Sesame, especially when the number of samples is very limited. Fig. 8(m)–(p) shows four visual results of the HC dataset, displaying varying degrees of misclassification for Bare Soil, Road, Bright Object, and Watermelon in all four visualized results, mainly due to the influence of shadows and the dense distribution of land cover types.

To demonstrate the effectiveness of the proposed multibranch framework, we individually generated and assessed the results of each branch in the network. Fig. 9 illustrates the classification accuracies of different branches on various datasets. The x-axis, y-axis, and z-axis of the coordinate system represent the different branches, datasets, and classification accuracies, respectively. It can be observed that the fusion of features at different scales yields optimal performance across all datasets, indicating the successful combination of features extracted from multiple scales for improved clustering results. Based on the results from other branches, larger convolutional kernels demonstrated optimal performance on the IP and PU datasets, likely because the IP dataset's higher

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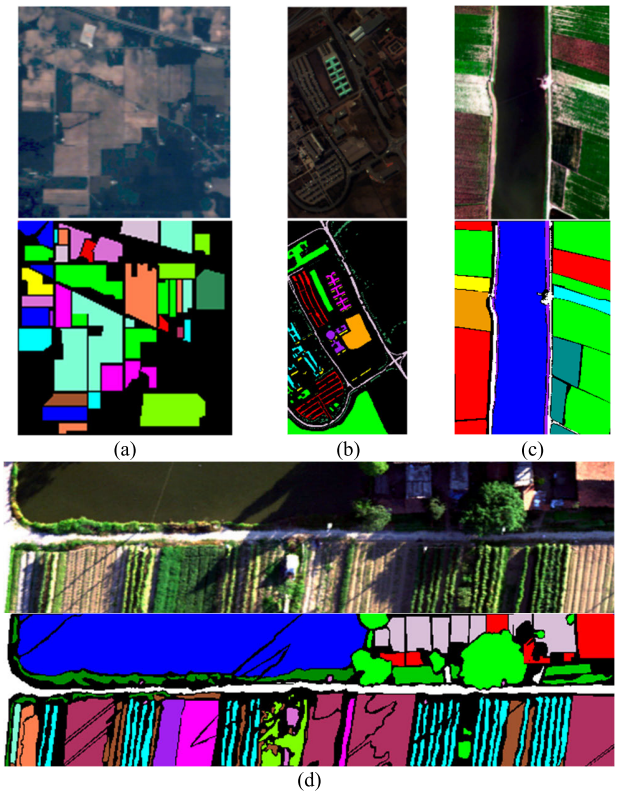


Fig. 7. False-color map and ground-truth map. (a) IP, (b) PU, (c) LK, and (d) HC.

spatial resolution enabled larger kernels to capture more spatial structural information, which in turn benefited performance on the PU dataset. Medium-sized kernels achieved optimal results on the LK dataset by striking a balance in receptive field size, thereby enhancing processing efficiency for datasets with evenly distributed features. Conversely, smaller kernels yielded superior results on the HC dataset, as excessively large kernels tend to introduce irrelevant information when handling complex feature distributions.

D. Ablation Experiments

As shown in Fig. 1, the proposed method comprises two attention mechanisms and one fusion strategy. In this section, ablation experiments were conducted to validate the effectiveness of these components. Table III displays the statistical results of the ablation experiments applied to the four datasets. The GDA module overcomes the limitations of convolutional networks in capturing long-range dependencies by integrating global information into the network.

Furthermore, it enhances the extraction of detailed information by focusing on the spectral features of central pixels. As listed in the table, preserving only the spectral attention module in our model leads to significant improvements compared to the experimental group without any modules across all four datasets. Specifically, for the IP dataset, the OA, AA, and Kappa improve by 0.59%, 0.64%, and 0.66%, respectively. For the HC dataset, the corresponding improvements in OA, AA, and Kappa are 1.13%, 1.66%, and 1.32%, respectively.

The DSA module adjusts the weights of each pixel by considering the spatial correlation between each pixel and its neighboring pixels, thereby minimizing the influence of heterogeneous pixels. The experimental group with the fusion strategy demonstrated significant improvements on all four datasets, compared to the experimental group without any modules. Specifically, for the LK dataset, the OA, AA, and Kappa improved by 1.07%, 2.50%, and 1.47%, respectively.

The AFF not only combines information from different scales and reduces redundant information generated after fusion, but also emphasizes the features that significantly influence the classification results. For the PU dataset compared to the experimental group without fusion strategy, the accuracy, average precision, and Kappa value rose by 0.41%, 0.34%, and 0.54% respectively.

In a multibranch network, each branch may handle distinct tasks or learn features across varied scales. Incorporating ALs accelerates network convergence and mitigates potential negative impacts of specific branches during training. In the HC dataset, the introduction of a fusion strategy led to improvements of 1.08% in accuracy, 1.42% in mean accuracy, and 1.25% in Kappa value compared to the control group. By incorporating all three modules into the classification method, optimal results were achieved in all the scenarios. Particularly, the OA has increased by 1.13%, 0.95%, 1.37%, and 1.33% on the IP, PU, LK, and HC datasets, respectively, indicating that the proposed modules do not interfere with each other.

E. Classification Performance

The proposed method achieves superior performance compared to the best-performing baselines for the IP dataset. We observed improvements of 0.91%, 2.28%, and 1.04% in terms of OA, AA, and Kappa metrics, respectively. Moreover, our method demonstrates smaller variances in these metrics, indicating that the accuracies for each class are relatively closer. In addition, our method exhibits advantageous handling of sample imbalance issues. Table IV highlights that our proposed method achieves the highest classification accuracy for low-sample classes, such as Alfalfa and Oats (classes with only 1 or 2 samples available). In contrast, multiscale classification methods of SSUN, EMFFN, and BTA-Net show poorer performance, likely due to the influence of spatial resolution of the dataset. In low-resolution scenarios, extracting beneficial multiscale features becomes challenging. Due to the absence of effective fusion strategies, Tri-CNN exhibits significant misclassification of land boundaries, potentially exacerbated by sample imbalance. It is noteworthy that our proposed efficient multiscale feature extraction network allows our approach to maintain high accuracy even with limited samples. Fig. 10 demonstrates that the proposed method performs well on Alfalfa, Grass-p, and Grass-p-m datasets, which contain few samples.

For the PU dataset, the proposed method surpasses the second-best method by Bricks exhibiting complex geometric shapes. Our fusion strategy effectively integrates the extracted geometric features, resulting in optimal performance for these two land covers 1.17%, 1.59%, and 1.56% in terms of OA, AA, and Kappa metrics, respectively. As listed in Table V, apart from Asphalt and Bricks, the classification results for

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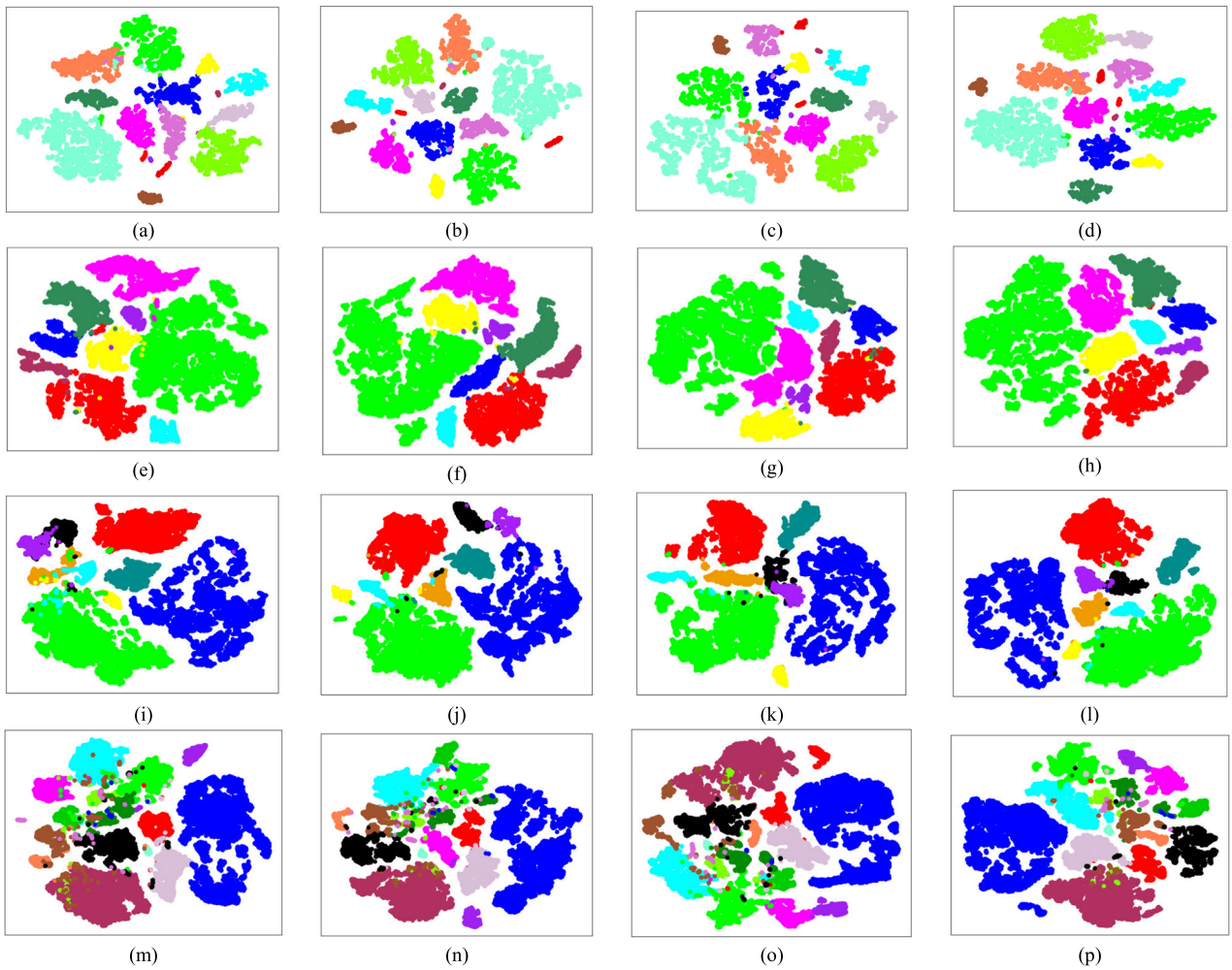


Fig. 8. Different scale classification feature visualization results. The first three columns correspond to different scale features extracted from three branches, and the fourth column is the visualization result after merging different scale features (the land cover types represented by colors are the same as Table II). Each line represents a different dataset. (a)–(d) IP, (e)–(h) PU, (i)–(l) LK, and (m)–(p) HC.

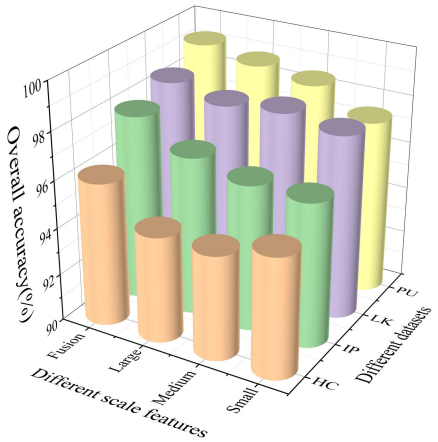


Fig. 9. Different feature classification results on different datasets.

other land covers consistently approach 99%. The land covers for Gravel and EMFFN attained the second-best results due to its robust capability in multiscale feature extraction. However, it exhibited intraclass misclassification on Bare Soil, characterized by a relatively uniform distribution pattern, possibly stemming from interference among different scale information during feature extraction. CSIL achieved the lowest performance, primarily due to the imbalance in training samples. As shown in Fig. 11, our proposed method demonstrates favorable performance in most land cover classifications, with minimal misclassifications.

For the LK dataset, the land covers primarily consist of crops and follow a regular distribution pattern. Most of the comparative methods achieved exceptional classification accuracies. As listed in Table VI, the proposed method surpasses the second-best method by 0.86%, 1.72%, and 1.12% in terms of OA, AA, and Kappa, respectively. Comparison with other methods, the proposed method achieves the highest accuracy in identifying narrow and elongated land covers such as roads, houses, and mixed weed on the LK dataset. This is attributed to DSA effectively mitigating the impact of intraclass variations on classification results. The MAFN method exhibits lower classification accuracy for the dataset due to its large parameter size and deep network and is more suitable for handling complex structured datasets and is prone to overfitting on datasets with regular spatial structures. Fig. 12 demonstrates that SSRN and FDSSC display significant misclassification for Sesame and Cotton land covers, which have similar spectral signatures. Employing multiscale information can assist the

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TABLE III
CLASSIFICATION PERFORMANCE OF THE ABLATION EXPERIMENTS (BOLD INDICATE THE BEST RESULT WITHIN THE SAME DATASET)

Dataset	GDA	DSA	AFF	AL	OA(%)	⬆	AA(%)	⬆	Kappa(%)	⬆
IP	✓	✓	✓	✓	**97.91±0.49**	**1.13**	**97.85±1.27**	**1.39**	**97.62±0.56**	**1.29**
	✓				97.37±0.31	0.59	97.10±1.36	0.64	96.99±0.36	0.66
		✓			96.98±0.44	0.20	97.21±1.09	0.75	96.56±0.51	0.23
			✓		97.01±0.48	0.23	97.14±0.97	0.68	96.58±0.55	0.25
				✓	97.18±0.35	0.30	97.46±0.61	1.00	96.79±0.40	0.46
					96.78±0.27	—	96.46±1.29	—	96.33±0.31	—
PU	✓	✓	✓	✓	**99.13±0.42**	**0.95**	**99.12±0.30**	**1.48**	**98.85±0.56**	**1.25**
	✓				99.00±0.24	0.81	98.59±0.38	0.95	98.67±0.32	1.07
		✓			98.74±0.33	0.55	98.38±0.63	0.74	98.33±0.44	0.73
			✓		98.60±0.44	0.41	97.98±0.59	0.34	98.14±0.58	0.54
				✓	98.85±0.59	0.66	98.56±049	0.92	98.48±0.79	0.88
					98.19±0.37	—	97.64±0.77	—	97.60±0.49	—
LK	✓	✓	✓	✓	**98.40±0.42**	**1.37**	**96.58±0.85**	**3.03**	**97.89±0.55**	**1.79**
	✓				97.40±0.59	0.37	94.00±1.51	0.45	96.57±0.78	0.47
		✓			98.10±0.24	1.07	96.05±1.15	2.50	97.57±3.14	1.47
			✓		97.24±0.53	0.21	94.25±0.97	0.70	96.36±0.70	0.26
				✓	97.11±0.44	0.08	93.98±1.09	0.43	96.20±0.58	0.10
					97.03±0.35	—	93.55±0.94	—	96.10±0.46	—
HC	✓	✓	✓	✓	**96.16±0.47**	**1.33**	**94.43±1.04**	**2.43**	**95.50±0.55**	**1.56**
	✓				95.96±0.29	1.13	93.66±0.74	1.66	95.26±0.34	1.32
		✓			95.02±1.23	0.19	92.86±1.29	0.86	94.16±1.44	0.22
			✓		95.54±0.35	0.71	92.93±1.12	0.93	94.77±0.41	0.83
				✓	95.91±0.53	1.08	93.46±1.42	1.46	95.19±0.62	1.25
					94.83±1.24	—	92.00±1.14	—	93.94±1.46	—

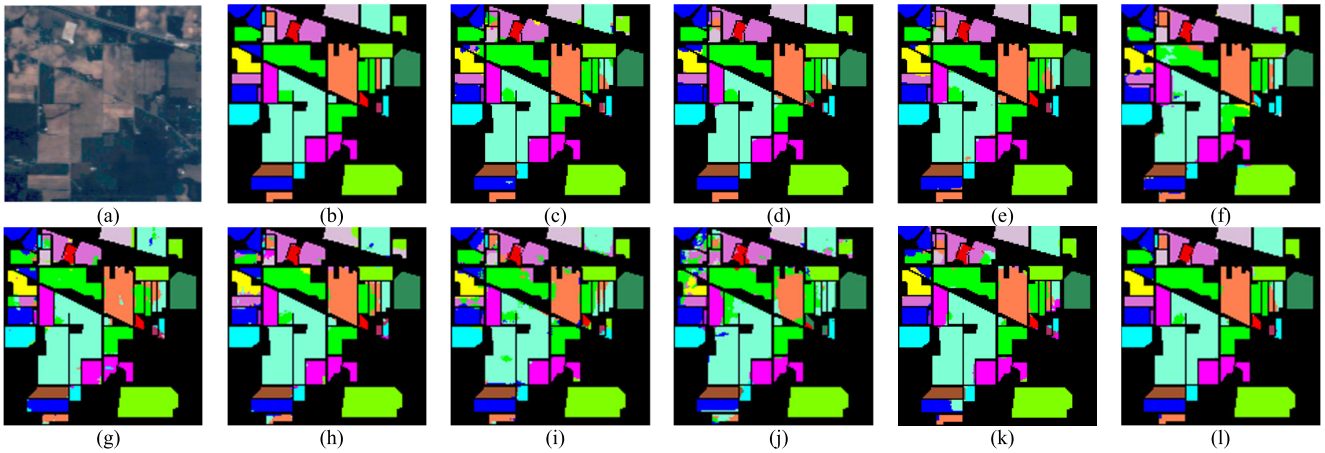


Fig. 10. Classification maps of different methods on the IP dataset. (a) Pseudo-color image, (b) ground-truth map, (c) SSRN, (d) FDSSC, (e) DBDAMN, (f) MAFN, (g) SSUN, (h) EMFFN, (i) BTA-Net, (j) Tri-CNN, (k) CSIL, and (l) MBAFFN (Ours).

model in distinguishing features with identical spectra more effectively.

For the HC dataset, certain land covers are affected by shadows, and the complex distribution of land covers increases the difficulty of classification. As listed in Table VII, our proposed method surpasses the second-best method by 3.16%, 6.89%, and 3.73% in terms of OA, AA, and Kappa, respectively. The proposed method demonstrates greater robustness when handling more complex datasets. Satisfactory results are achieved for uniformly distributed land cover categories such as Cowpea, Strawberry, and Sorghum. The BTA-Net employs multiscale Bole convolution for feature extraction, but this method sacrifices neighboring information crucial for classification, thereby leading to inferior performance when dealing

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Class	SSRN	FDSSC	DBDAMN	MAFN	SSUN	EMFFN	BTA-Net	Tri-CNN	CSIL	MBAFFN (Ours)
1	93.28±8.57	80.00±40.00	98.33±4.27	89.44±13.32	98.83±1.86	94.67±8.44	94.71±5.52	60.95±25.67	88.86±10.19	**100±0**
2	95.74±2.14	**97.40±1.23**	94.72±3.55	93.74±4.42	90.66±3.10	96.26±2.26	91.62±1.79	78.32±6.23	90.77±3.60	96.83±1.43
3	96.13±1.83	94.93±2.31	95.65±2.98	90.18±3.79	90.97±3.16	91.98±5.68	87.97±3.70	78.12±8.95	88.96±4.93	**97.63±0.95**
4	95.52±3.57	98.10±2.46	95.68±7.00	95.20±5.06	92.09±6.45	89.32±8.71	89.74±4.55	52.62±10.63	93.11±5.78	**98.39±2.53**
5	**98.96±1.36**	96.72±2.59	**98.96±0.98**	95.42±3.44	97.02±2.19	97.37±3.02	93.81±5.31	89.59±7.35	88.58±3.71	97.76±1.36
6	99.14±0.84	97.82±1.41	98.88±1.11	98.09±1.28	96.51±1.70	96.33±1.84	98.46±1.16	98.19±1.58	97.42±2.42	**99.45±0.75**
7	92.01±1.64	76.78±30.18	76.15±19.12	69.59±19.07	78.53±17.68	78.14±19.56	86.37±14.16	32.31±12.81	**89.62±11.15**	88.54±17.68
8	99.33±1.15	96.15±2.50	**100.00±0.00**	97.28±4.83	99.74±0.29	97.39±2.19	97.11±1.87	99.98±0.07	99.96±0.09	99.47±0.88
9	82.91±29.27	20.00±40.00	95.35±7.32	88.12±16.48	93.96±7.25	84.37±29.05	91.80±12.13	29.44±23.71	43.16±21.29	**100±0**
10	92.68±1.92	94.26±2.29	93.93±2.38	92.70±2.43	90.07±3.48	93.33±2.61	91.70±3.75	83.11±5.27	92.48±4.01	**96.78±1.54**
11	97.37±0.81	**98.34±1.03**	**98.34±1.03**	97.04±2.35	95.00±0.93	94.69±2.90	93.90±1.35	90.43±5.07	97.28±1.15	98.13±1.04
12	95.68±2.66	94.85±3.00	96.18±2.88	92.03±3.26	89.26±3.98	89.83±5.49	90.87±4.32	72.39±9.62	87.73±3.66	**96.94±1.85**
13	99.78±0.50	99.89±0.22	98.63±2.25	96.54±2.63	96.42±3.47	96.29±4.68	99.26±1.42	97.03±2.32	96.41±3.44	**99.95±0.16**
14	97.60±0.85	98.07±1.29	98.44±1.23	**99.16±1.37**	97.88±1.40	98.14±1.90	97.37±1.43	98.17±1.88	98.75±0.86	98.57±0.93
15	96.79±2.47	97.99±1.88	94.87±4.02	92.43±5.31	91.79±4.03	92.07±4.45	92.72±3.83	74.13±13.94	96.80±2.66	**98.81±1.26**
16	96.20±4.19	**99.39±0.99**	93.97±5.94	94.48±6.85	96.21±1.45	95.19±6.16	90.38±5.51	88.10±10.05	88.75±11.98	98.32±1.89
OA (%)	96.67±0.60	97.00±0.79	96.79±0.74	95.01±1.50	93.76±0.60	94.60±0.96	93.41±0.73	86.12±1.86	94.23±0.91	**97.91±0.49**
AA (%)	95.57±2.23	90.04±5.38	95.51±1.04	92.59±2.15	93.43±1.19	92.84±1.98	92.99±1.82	76.43±3.47	89.91±2.06	**97.85±1.27**
Kappa (%)	96.21±0.69	91.61±1.35	96.58±0.90	96.35±0.84	92.89±0.69	93.83±1.10	92.48±0.83	84.13±2.12	93.42±1.04	**97.62±0.56**

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Class	SSRN	FDSSC	DBDAMN	MAFN	SSUN	EMFFN	BTA-Net	Tri-CNN	CSIL	MBAFFN (Ours)
1	97.59±2.45	96.71±4.40	97.92±1.07	96.98±1.29	93.33±2.01	97.63±1.53	93.90±3.46	93.63±3.67	92.70±3.25	**98.93±0.71**
2	98.72±0.73	99.56±0.33	**99.58±0.16**	99.31±0.34	96.16±1.41	98.83±0.75	99.05±0.83	99.39±1.00	98.98±0.38	99.41±0.72
3	89.87±12.05	97.83±3.13	97.22±6.14	93.79±4.20	85.20±3.52	93.80±5.75	94.91±4.64	81.17±9.43	90.26±5.01	**98.73±1.94**
4	99.51±0.59	98.49±0.98	98.17±0.71	99.03±0.82	98.18±1.98	99.49±0.35	**99.62±0.37**	91.66±5.41	85.55±6.54	99.29±0.68
5	99.96±0.04	99.11±1.21	99.50±0.75	99.05±1.76	97.86±2.37	99.02±1.80	99.72±0.35	**100±0.00**	98.27±1.82	99.84±0.16
6	97.73±1.58	99.32±1.06	98.26±2.53	97.98±1.48	95.15±1.94	99..27±0.54	93.14±10.23	91.72±6.33	98.43±0.85	**99.83±0.30**
7	97.41±4.81	**99.83±0.47**	96.74±8.40	95.88±2.53	88.33±4.85	95.13±6.67	89.90±13.41	93.05±5.65	86.51±5.52	99.72±0.43
8	89.16±8.02	87.86±8.42	89.80±4.56	94.02±2.25	89.00±3.33	94.68±4.39	84.17±7.29	87.07±6.91	79.41±6.44	**96.87±1.93**
9	99.87±0.32	99.05±1.14	98.36±1.31	98.68±1.68	98.34±2.95	99.53±1.13	**99.94±0.11**	95.26±3.04	56.98±16.68	99.48±0.55
OA (%)	96.94±1.52	97.64±1.87	97.82±0.82	97.90±051	94.43±1.03	97.96±0.74	95.17±1.68	94.82±1.16	93.53±1.15	**99.13±0.42**
AA (%)	96.65±1.67	97.53±1.40	97.29±1.48	97.19±0.82	93.51±1.09	97.49±1.23	94.92±1.35	92.55±1.51	87.46±2.69	**99.12±0.30**
Kappa (%)	95.93±2.03	96.86±2.50	97.12±1.08	97.21±0.68	92.58±1.40	97.29±0.99	93.63±2.16	93.09±1.56	91.40±1.55	**98.85±0.56**

with complex land cover distributions. Fig. 13 clearly shows that the proposed method can minimize misclassification to a significant extent in the intricate scenarios of mixed distribution of multiple land covers, such as Water spinach, Watermelon, and Bare soil, as well as in the presence of shadows.

F. Loss Coefficient Experiment

This section presents experimental findings on the influence of varying coefficients for ALs and the impact of omitting these losses on experimental outcomes. Table VIII details specific results across four datasets. The experiments showed that setting the loss coefficients uniformly to 0.25 resulted in optimal performance across all datasets. Specifically, compared to scenarios that exclude ALs, OA increased by 2.11%, 0.04%, 0.36%, and 1.15% across the respective datasets. The improvements for the PU and LK datasets were relatively minor, possibly due to higher quality training samples, which resulted in lower model training losses, thereby lessening the impact of the coefficients. Cases II and III examined the effects of strengthening or weakening ALs during model training, both yielding inferior results compared to Case I. This suggests that uniform coefficients can improve network stability and

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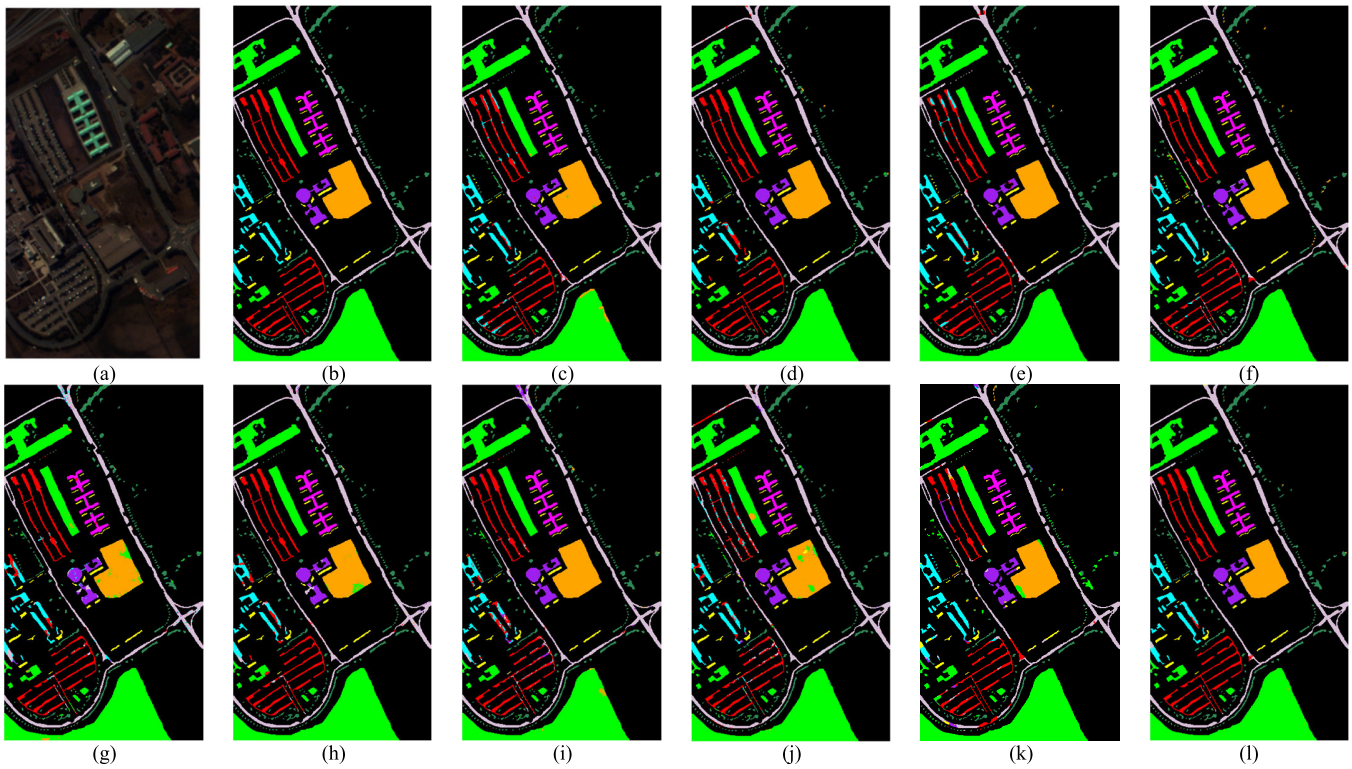


Fig. 11. Classification maps of different methods on the PU dataset. (a) Pseudo-color image, (b) ground-truth map, (c) SSRN, (d) FDSSC, (e) DBDAMN, (f) MAFN, (g) SSUN, (h) EMFFN, (i) BTA-Net, (j) Tri-CNN, (k) CSIL, and (l) MBAFFN (Ours).

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE LK DATASET (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Class	SSRN	FDSSC	DBDAMN	MAFN	SSUN	EMFFN	BTA-Net	Tri-CNN	CSIL	MBAFFN (Ours)
1	99.12±0.86	99.03±0.77	99.37±0.55	90.85±7.18	99.03±1.19	98.52±1.27	98.73±0.91	99.09±0.54	96.92±2.72	**99.49±0.27**
2	88.12±14.15	84.66±15.22	93.92±5.26	45.37±19.97	88.08±7.67	91.40±2.53	93.58±3.38	82.39±12.25	92.43±4.96	**98.35±0.96**
3	97.10±3.35	98.64±2.30	96.37±2.73	45.05±39.35	91.67±7.27	89.98±6.83	97.49±4.65	94.83±6.01	**99.37±0.67**	97.77±2.94
4	96.58±1.26	94.61±2.77	97.26±1.54	91.85±6.03	95.79±0.98	98.12±0.60	90.93±3.08	97.86±1.17	97.46±1.11	**98.47±0.54**
5	84.90±29.00	79.31±39.68	96.04±4.07	37.10±36.19	83.63±13.21	86.84±7.90	88.57±12.50	75.75±14.74	81.92±10.05	**98.01±2.98**
6	94.06±8.33	94.46±3.64	98.02±1.70	84.01±9.64	97.15±1.35	96.81±2.42	98.25±2.38	97.51±3.44	90.17±4.44	**98.99±1.04**
7	99.18±0.63	97.07±1.16	99.14±0.67	98.55±1.14	99.36±0.41	**99.91±0.11**	99.37±0.42	99.26±0.73	99.74±0.28	99.62±0.52
8	89.27±7.38	73.61±10.66	80.71±9.95	60.99±22.95	**95.47±4.42**	90.58±6.88	81.75±11.36	76.91±9.09	56.57±9.33	89.84±6.32
9	88.87±2.08	64.78±43.26	79.15±10.46	93.48±16.34	**97.80±1.50**	89.55±8.80	86.43±8.08	69.31±15.08	71.18±5.27	88.70±7.98
OA (%)	96.64±1.40	94.64±1.80	96.91±0.68	88.89±1.82	96.91±0.57	97.54±0.69	92.79±1.80	95.93±0.95	95.12±0.56	**98.40±0.42**
AA (%)	93.02±3.83	87.35±10.17	93.33±1.01	71.92±9.47	94.22±1.31	93.52±2.26	94.86±0.86	88.10±3.05	87.31±1.38	**96.58±0.85**
Kappa (%)	95.55±1.85	92.87±2.42	95.93±0.91	85.27±2.50	95.93±0.74	96.77±0.90	93.17±1.16	94.64±1.25	93.57±0.74	**97.89±0.55**

accelerate convergence. Significant disparities in coefficients may lead certain branches to converge too quickly or too slowly, thereby affecting overall network learning efficiency and stability. Consequently, Case I was selected as the coefficient setting for ALs in our experiments.

G. Influence of Model Parameters

The proper selection of the sample size is important when training neural networks. In this section, we further investigate the impact of sample quantity on all the methods. Fig. 14 presents the classification results of all the methods with sample sizes on various datasets. It can be observed that our approach consistently achieves optimal results across the four datasets. On the IP dataset, the classification accuracy of each method improves gradually as the number of training samples increases. This trend is primarily attributed to the dataset's low spatial resolution, which constrains the extraction of spatial features and necessitates additional training samples for compensation. For the PU dataset, most classification methods initially demonstrate significant accuracy improvements with increasing sample sizes. Notably, FDSSC, DBDAMN, and MBAFFN also exhibit satisfactory accuracy even with small sample sizes. The first two methods benefit from their deeper

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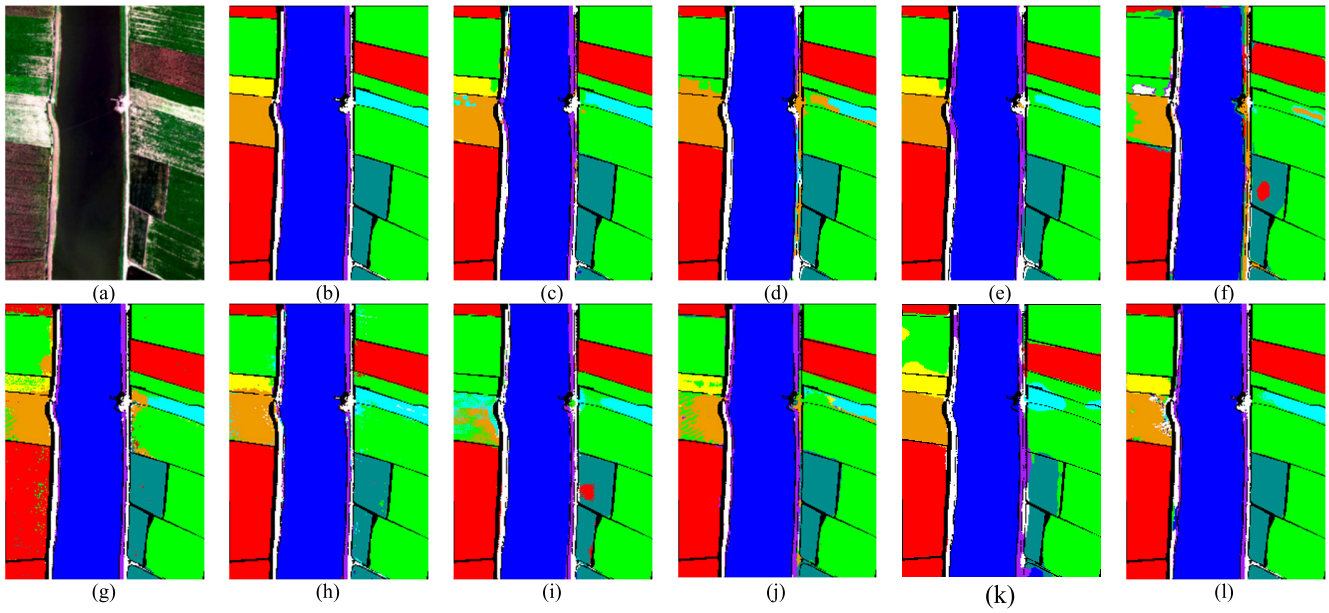


Fig. 12. Classification maps of different methods on the LK dataset. (a) Pseudo-color image, (b) ground-truth map, (c) SSRN, (d) FDSSC, (e) DBDAMN, (f) MAFN, (g) SSUN, (h) EMFFN, (i) BTA-Net, (j) Tri-CNN, (k) CSIL, and (l) MBAFFN (Ours).

TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HC DATASET (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Class	SSRN	FDSSC	DBDAMN	MAFN	SSUN	EMFFN	BTA-Net	Tri-CNN	CSIL	MBAFFN (Ours)
1	90.72±2.56	92.45±3.71	94.38±1.17	**97.5±1.16**	91.87±1.28	95.62±2.10	88.90±2.25	97.79±1.67	96.76±1.40	96.79±0.87
2	87.11±2.78	85.76±7.31	88.07±7.75	93.26±3.66	87.72±3.07	95.22±1.73	87.08±4.70	91.16±4.11	90.95±4.27	**95.11±2.01**
3	88.38±4.60	90.70±4.20	91.22±7.27	88.86±5.32	84.01±6.74	89.45±6.00	80.19±7.45	83.76±9.91	90.48±6.78	**95.02±3.62**
4	95.29±3.45	**98.83±4.20**	95.69±6.92	92.13±5.69	90.15±6.47	84.49±7.67	93.67±4.49	94.70±5.63	89.69±5.02	97.82±3.62
5	56.26±46.90	20.00±40.00	57.83±47.50	65.23±42.92	77.01±12.13	66.94±15.16	85.58±9.13	43.61±35.62	89.53±7.31	**92.07±7.79**
6	61.98±25.84	69.19±36.70	82.25±11.91	68.98±10.26	57.41±10.82	76.62±13.14	69.29±10.04	39.52±21.25	73.54±11.43	**83.03±9.07**
7	79.22±5.05	86.41±4.46	82.08±10.37	74.63±10.83	79.15±8.67	83.80±9.99	78.45±7.33	76.89±6.85	71.30±10.36	**96.59±2.10**
8	87.83±6.48	92.88±6.07	92.16±4.36	90.79±4.16	86.78±3.00	93.26±2.53	82.36±2.91	70.28±7.20	89.15±2.16	**93.31±2.89**
9	84.13±5.23	82.31±4.88	83.59±7.38	77.95±9.71	82.28±12.39	87.78±6.38	82.11±5.72	59.93±15.02	70.76±4.89	**90.83±5.50**
10	97.09±1.77	**98.92±0.43**	97.75±3.66	92.57±4.01	93.49±3.98	89.14±9.23	87.89±6.32	82.49±9.15	82.45±6.21	98.80±0.73
11	92.72±2.28	91.47±6.37	93.11±2.78	89.28±6.85	86.81±3.32	87.61±3.37	89.80±3.96	87.08±7.91	89.71±6.25	**95.44±1.84**
12	73.56±37.22	85.04±28.61	89.60±24.45	78.41±10.0	79.75±8.15	89.41±8.01	71.32±10.66	38.50±17.25	94.33±6.35	**95.06±3.63**
13	75.57±6.09	76.91±16.26	72.15±11.20	82.21±6.22	68.54±7.36	84.19±9.99	76.91±7.63	56.15±11.26	68.86±7.39	**86.49±5.31**
14	83.66±2.72	83.39±3.05	83.12±5.94	87.64±4.32	90.26±3.38	94.41±2.02	85.84±6.32	84.80±8.34	81.69±4.65	**94.80±2.05**
15	89.46±29.86	80.00±40.00	97.75±3.96	88.24±10.95	82.51±12.39	76.53±17.69	95.83±4.97	18.43±11.89	65.28±18.93	**100±0**
16	99.60±0.48	99.68±0.20	99.52±0.49	99.28±0.57	98.51±5.43	97.81±1.74	98.84±0.81	99.19±1.36	98.38±0.69	**99.68±0.16**
OA (%)	91.23±1.00	92.35±2.52	92.56±2.63	92.31±1.07	89.82±0.51	93.00±0.82	89.17±0.86	87.00±1.38	90.54±1.20	**96.16±0.47**
AA (%)	83.91±7.53	83.37±8.57	87.54±6.37	85.44±4.00	83.51±1.42	87.02±2.16	84.62±2.14	70.27±3.74	83.93±2.21	**94.43±1.04**
Kappa (%)	89.69±1.19	91.01±2.98	91.26±3.09	90.99±1.25	88.08±0.64	91.77±0.96	87.28±1.00	84.50±1.64	88.89±1.42	**95.50±0.55**

network architectures, which are capable of capturing more profound semantic information. Our proposed method achieves high classification accuracy under low-sample conditions by integrating multiscale information. On the LK dataset, while most methods show stable performance reflecting dataset characteristics, DBDAMN and FDSSC experience reduced OA when the sample ratio exceeds 1.5%, likely due to overfitting caused by their deeper network architectures during training. Conversely, on the HC dataset, all methods demonstrate enhanced classification accuracy with increasing sample ratios. MBAFFN and EMFFN particularly excel on datasets featuring complex land cover distributions, leveraging their superior multiscale feature extraction capabilities.

After formulating the MBAFFN framework, we proceed to investigate the optimal number of convolutional kernels for each convolutional layer in the spatial feature extraction

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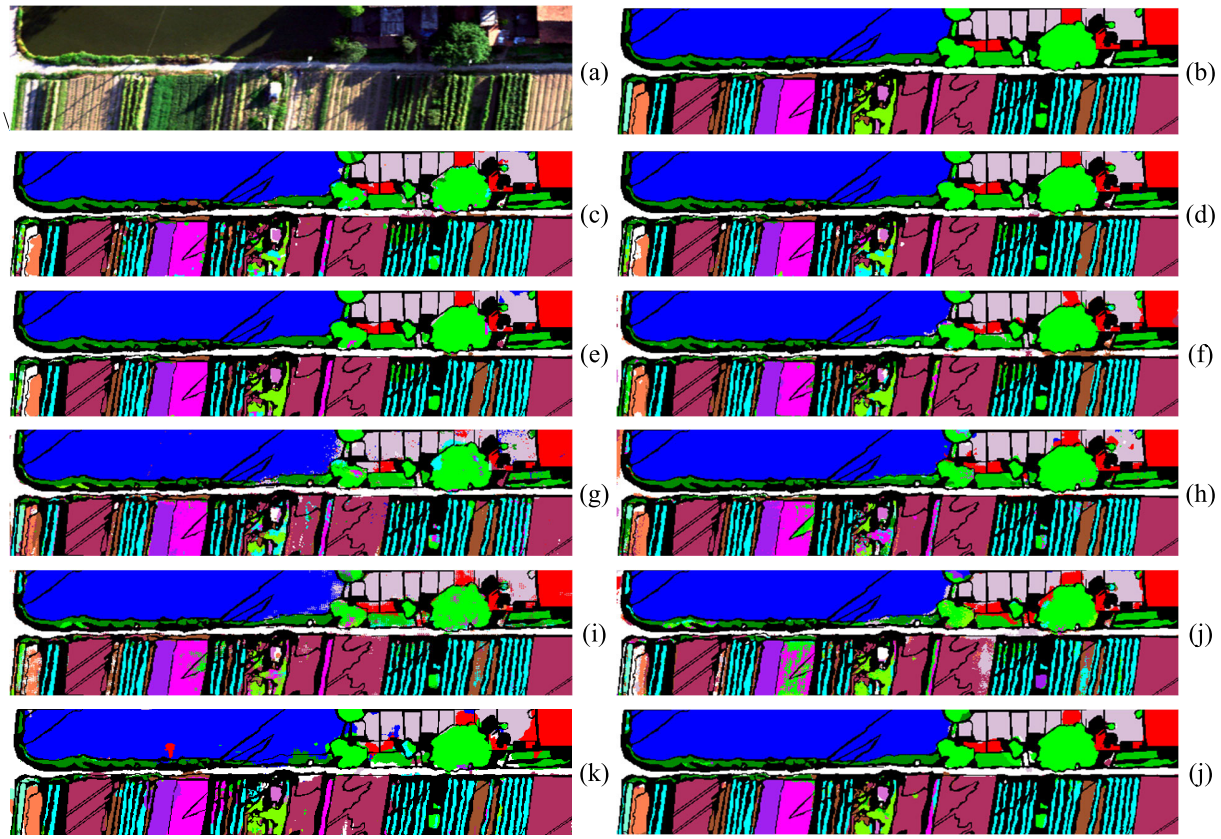


Fig. 13. Classification maps of different methods on the HC dataset. (a) Pseudo-color image, (b) ground-truth map, (c) SSRN, (d) FDSSC, (e) DBDAMN, (f) MAFN, (g) SSUN, (h) EMFFN, (i) BTA-Net, (j) Tri-CNN, (k) CSIL, and (l) MBAFFN (Ours).

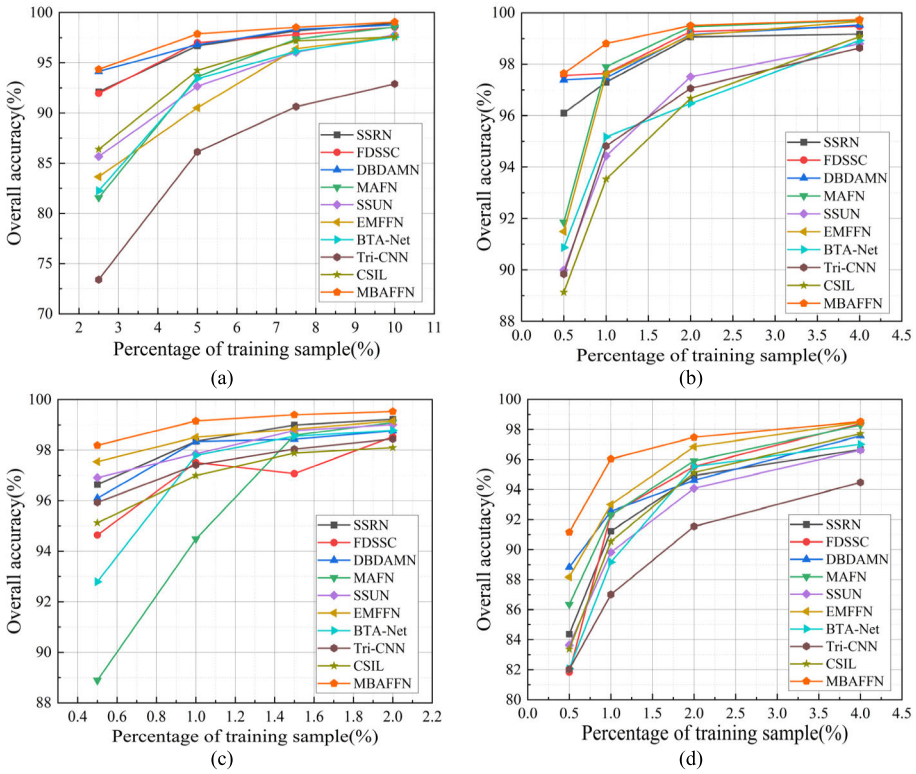


Fig. 14. OA of different models with different training samples' percentages. (a) IP, (b) PU, (c) LK, and (d) HC.

network. In the spectral feature extraction stage, the proposed network employs the same number of convolutions as the kernel size. However, during the spatial feature extraction, the number of convolutional kernels affects both the representation

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TABLE VIII
RESULTS OF DIFFERENT LOSS COEFFICIENTS ON FOUR DATASETS (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Dataset	Case	OA (%)	AA (%)	Kappa (%)
IP	I	**97.91±0.49**	**97.85±1.27**	**97.62±0.56**
	II	97.05±1.78	96.02±4.67	96.64±2.03
	III	97.55±0.94	97.69±1.12	97.21±1.01
	IV	95.80±3.29	96.91±1.94	95.21±3.74
PU	I	**99.13±0.42**	**99.12±0.30**	**98.85±0.56**
	II	99.04±0.30	98.93±0.40	98.73±0.40
	III	99.10±0.45	98.72±0.78	98.81±0.30
	IV	99.09±0.22	98.90±0.43	98.80±0.30
LK	I	**98.40±0.42**	**96.58±0.85**	**97.89±0.55**
	II	98.25±0.40	96.55±0.56	97.70±0.52
	III	98.19±0.60	95.95±1.24	97.62±0.79
	IV	98.04±0.49	95.34±1.41	97.42±0.65
HC	I	**96.16±0.47**	**94.43±1.04**	**95.50±0.55**
	II	95.99±0.58	94.10±1.12	95.30±0.68
	III	95.81±0.43	93.94±1.04	95.09±0.50
	IV	95.01±0.71	92.40±1.45	94.15±0.83

ANNOTATION: In case I, $\lambda^0, \lambda^1, , \lambda^2$ and λ^3 are all set to 0.25. In case II, $\lambda^0, \lambda^1, , \lambda^2$ and λ^3 are respectively set to 0.3, 0.3, 0.3, and 0.1. In case III, $\lambda^0, \lambda^1, , \lambda^2$ and λ^3 are respectively set to 0.1, 0.1, 0.1, and 0.7. In case IV, λ^0 is set to 1, while the others are set to 0.

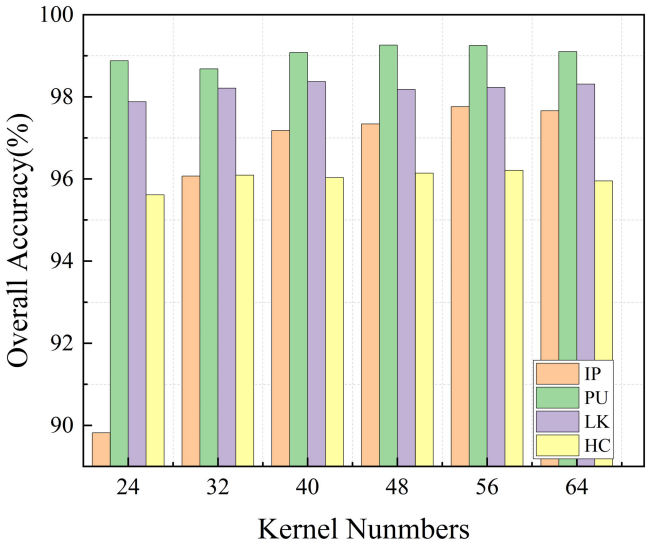


Fig. 15. OA of MBAFFN with different kernel numbers in the IP, PU, LK, and HC datasets.

capability and computational cost of MBAFFN. The impact of different numbers of convolutional kernels on the classification results is shown in Fig. 15. It can be observed that we evaluate various numbers of kernels ranging from 24 to 64 with intervals of 8, in each convolutional layer. MBAFFN demonstrates the highest OA of 97.91% for the IP dataset with a kernel size of 56. The 40-kernel model obtains the highest OA of 98.40% for the LK dataset, while the 48-kernel model yields the best performance for the PU and HC datasets with OA values of 99.13% and 96.16%, respectively.

Patch size is another important hyperparameter for deep learning models. To analyze its impact on classification results using the proposed method, we employed five patch sizes of 7 × 7, 9 × 9, 11 × 11, 13 × 13, and 15 × 15 as model parameters. Fig. 16 presents the classification results of

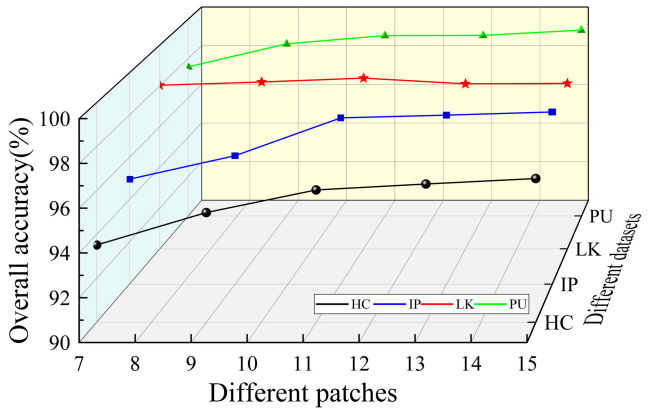


Fig. 16. Overall accuracies with different patch size for three experimental datasets.

different patch sizes for different datasets. It can be observed that the OA is 96.16% and 97.91% for the HC and IP datasets when the patch size is 11, respectively. As the patch size increases, the improvement in accuracy becomes less significant, since the HC dataset has a complex distribution of objects, and larger patch sizes may introduce more interference information. For the IP dataset, with its lower spatial resolution, larger patch sizes might reduce the network's focus on the center pixel. For the PU dataset, classification accuracy continues to improve with larger patch sizes, with the highest OA of 99.45% achieved when the patch size is 15. The complex distribution of objects in the PU dataset benefits from larger patch sizes, which provide multiscale structural information and improve classification accuracy. For the LK dataset, the maximum OA is 98.40% when the patch size is 11, and the accuracy slightly decreases as the patch size increases. This is because this dataset has a relatively simple spatial distribution and a small number of training samples (only 0.5%), and larger patches at object edges are more susceptible to interference from heterogeneous pixels. Considering the computational resources required, we decided to use a patch size of 11 for all experiments.

H. Computational Efficiency

In this section, we analyze the computational efficiency of various classification methods. Table IX summarizes the training time, testing time, total parameters, and computational complexity for the IP, PU, LK, and HC datasets. MBAFFN exhibited commendable performance in training time, consistently ranking among the top three across all datasets. This can be attributed to its effective multiscale feature extraction method, facilitating quicker model convergence and reduced training duration. SSUN demonstrated the best training time across the IP, LK, and HC datasets, leveraging fewer model parameters and lower computational complexity, primarily through the use of 2-D convolutional layers and maintaining a relatively shallow model depth. In contrast, both EMFFN and CSIL showed prolonged training times. The former incorporated numerous serial structures within the multibranch network, increasing computational load and interfering with feature extraction across different scales. The latter experienced heightened computational complexity due to multilevel

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TABLE IX
COMPUTATIONAL TIME (I.E., TRAINING TIME AND TEST TIME) AND TOTAL NUMBER OF TRAINING PARAMETERS FOR DEEP LEARNING METHODS (BOLD INDICATE THE BEST RESULT FOR EACH LINE)

Method		SSRN	FDSSC	DBDAMN	MAFN	SSUN	EMFFN	BTA-Net	Tri-CNN	CSIL	MBAFFN(Ours)
IP	Train/s	100.33	221.69	104.88	208.18	**82.20**	643.88	112.29	99.04	1142.36	92.9
	Test/s	5.69	4.68	7.16	8.14	3.29	15.13	**1.67**	3.24	1094.74	2.91
	Params/M	**0.36**	1.23	0.38	5.57	0.95	831.65	1.19	322.11	132.43	1.11
	FLoPs/M	158.38	177.40	107.98	604.11	**6.36**	868.95	71.93	1979.06	148326.20	140.41
	FPS	35.15	42.74	27.93	28.26	62.01	13.54	5.98	13.89	0.005	**68.04**
PU	Train/s	84.33	106.41	71.34	173.65	39.5	679.38	123.99	25.74	2567.88	**25.18**
	Test/s	30.32	17.87	35.67	27.48	5.14	12.17	**1.43**	9.57	2657.22	7.99
	Params/M	0.22	0.65	**0.20**	4.97	0.93	458.86	1.18	130.22	127.92	0.72
	FLoPs/M	81.21	91.79	68.98	531.80	**6.35**	491.44	86.60	601.44	148191.81	87.62
	FPS	4.00	5.76	5.56	4.84	**20.82**	8.87	6.99	4.70	0.002	12.88
LK	Train/s	56.30	163.06	70.93	134.28	25.7	652.16	85.46	**21.96**	976.09	33.99
	Test/s	37.27	33.33	42.48	43.58	6.05	27.43	**2.38**	9.58	1919.4	8.54
	Params/M	**0.47**	1.65	0.51	6.00	0.96	1100.33	1.18	50.79	124.94	0.51
	FLoPs/M	214.65	239.83	146.41	656.81	**6.35**	1141.04	86.60	215.34	148102.23	63.95
	FPS	7.24	8.10	6.36	6.88	**42.15**	10.03	4.20	4.70	0.003	10.54
HC	Train/s	132.65	421.97	185.64	275.97	**56.97**	1287.0	185.04	57.19	3856.04	61.52
	Test/s	45.42	42.88	60.63	53.80	7.61	30.77	**1.67**	13.95	3889.86	15.13
	Params/M	**0.48**	1.67	0.52	6.03	0.96	1115.96	1.19	85.66	124.96	0.71
	FLoPs/M	217.87	243.40	148.61	659.83	**6.35**	1156.86	71.93	383.17	148102.25	87.62
	FPS	6.02	6.39	4.52	5.65	**36.53**	9.07	5.99	3.23	0.002	5.95

input data, aimed at capturing multiscale information but ultimately compromising computational efficiency. Despite its higher computational complexity, the Tri-CNN method achieved the best and second-best training time results across all datasets. The extensive utilization of 3-D convolutions in Tri-CNN has led to notable overfitting issues. Moreover, the inadequate integration of results from multiple branches has introduced noise, thereby exerting a detrimental influence on the final classification outcomes. Computational benchmarks such as FLOPS and FPS confirm that MBAFFN, despite its higher computational complexity compared to methods like DBDAMN, SSUN, and BTA-Net, achieves superior computational efficiency. This success can be attributed to our exceptional parallel framework featuring robust multibranch architectures.

IV. CONCLUSION

In this article, we present a novel MBAFFN for hyperspectral remote sensing image classification. The framework comprises three parallel multiscale feature extraction networks to mitigate interference between different feature types. In the feature extraction stage, we overcome the limitation of CNNs in capturing global features and emphasize the significance of central pixels in the classification results. Meanwhile, we develop two attention modules to enhance the network's capabilities. Furthermore, we design a point-based feature fusion module that integrates results from multiple branches based on the importance of individual pixels within the same dimension. Our method improves the OA by 0.91%, 1.71%, 0.86%, and 3.16% on the IP, PU, LK, and HC datasets, respectively, outperforming traditional methods and state-of-the-art methods under conditions of imbalanced and small sample sizes. In the future, we aim to make full use of unlabeled data and further reduce the reliance on labeled data by employing unsupervised learning.

ACKNOWLEDGMENT

The authors would like to thank the handling editor and three anonymous reviewers for their valuable comments and suggestions, which significantly improved the quality of this article.

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